AI202 Artificial Intelligence
Lab Assignment 1

U24AI025

Q1 Consider a map given in image.

Implement Breadth First Search(BFS) and Depth First Search (DFS) algorithm to find all possible step cost between Syracuse to Chicago.

```
#include<bits/stdc++.h>
using namespace std;


vector<vector<pair<int,int>>> graph(14);
string city[14] = {
    "Chicago","Indianapolis","Columbus","Cleveland","Detroit",
    "Buffalo","Pittsburgh","Syracuse","Baltimore","Philadelphia",
    "New York","Providence","Boston","Portland"
};

void add_edge(int u, int v, int w){
    graph[u].push_back({v,w});
    graph[v].push_back({u,w});
}
```

```
void make_graph() {
    add_edge(0,4,283); add_edge(0,3,345); add_edge(0,1,182);
    add_edge(1,2,176);
    add_edge(2,3,144); add_edge(2,6,185);
    add_edge(3,4,169); add_edge(3,5,189); add_edge(3,6,134);
    add_edge(4,5,256);
    add_edge(5,6,215); add_edge(5,7,150);
    add_edge(6,8,247); add_edge(6,9,305);
    add_edge(7,10,254); add_edge(7,12,312); add_edge(7,9,253);
    add_edge(8,9,101);
    add_edge(9,10,97);
    add_edge(10,11,181); add_edge(10,12,215);
    add_edge(11,12,50);
    add_edge(12,13,107);
}

int city_to_index(string name){
    for(int i=0;i<14;i++){
        if(city[i] == name)
            return i;
    }
    return -1;
}


void dfs(int u,int d,
         vector<vector<pair<int,int>>>& g,
         vector<int>& vis,
         vector<int>& path,
         int cost)
{
    vis[u]=1;
    path.push_back(u);

    if(u==d){
        for(int x:path) cout<<city[x]<<" -> ";
        cout<<"Cost = "<<cost<<"\n";
    } else {
        for(auto e:g[u]){
            if(!vis[e.first])
```

```
                dfs(e.first,d,g,vis,path,cost+e.second);
        }
    }

    path.pop_back();
    vis[u]=0;
}


void bfs(int s,int d,vector<vector<pair<int,int>>>& g)
{
    vector<vector<int>> qpath;
    vector<int> qnode, qcost;

    qpath.push_back({s});
    qnode.push_back(s);
    qcost.push_back(0);

    int idx=0;
    while(idx<qnode.size()){
        int u=qnode[idx];
        int cost=qcost[idx];
        vector<int> path=qpath[idx];
        idx++;

        if(u==d){
            for(int x:path) cout<<city[x]<<" -> ";
            cout<<"Cost = "<<cost<<"\n";
            continue;
        }

        for(auto e:g[u]){
            bool used=false;
            for(int x:path) if(x==e.first) used=true;
            if(!used){
                vector<int> np=path;
                np.push_back(e.first);
                qpath.push_back(np);
                qnode.push_back(e.first);
                qcost.push_back(cost+e.second);
            }
```

```
        }
    }
}


int bfs_exploration_cost(int s,int d,
                         vector<vector<pair<int,int>>>& g)
{
    vector<vector<int>> qpath;
    vector<int> qnode, qcost;

    qpath.push_back({s});
    qnode.push_back(s);
    qcost.push_back(0);

    int idx=0, total=0;

    while(idx<qnode.size()){
        int u=qnode[idx];
        int cost=qcost[idx];
        vector<int> path=qpath[idx];
        idx++;

        total += cost;

        if(u==d) break;

        for(auto e:g[u]){
            bool used=false;
            for(int x:path) if(x==e.first) used=true;
            if(!used){
                vector<int> np=path;
                np.push_back(e.first);
                qpath.push_back(np);
                qnode.push_back(e.first);
                qcost.push_back(
                    e.second);
            }
        }
    }
    return total;
```

```
}

int main(){
    make_graph();

    string startCity, endCity;
    cin >> startCity >> endCity;

    int s = city_to_index(startCity);
    int d = city_to_index(endCity);

    if(s == -1 || d == -1){
        cout << "Invalid city name\n";
        return 0;
    }

    vector<int> visited(14,0), path;

    cout << "DFS paths:\n";
    dfs(s, d, graph, visited, path, 0);

    cout << "\nBFS paths:\n";
    bfs(s, d, graph);

    int exp_cost = bfs_exploration_cost(s, d, graph);
    cout << "\nBFS Exploration Cost = " << exp_cost << "\n";

    return 0;
}
```

DFS paths:

Syracuse -> Buffalo -> Cleveland -> Chicago -> Cost = 684
Syracuse -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 841
Syracuse -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 791
Syracuse -> Buffalo -> Cleveland -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1016
Syracuse -> Buffalo -> Detroit -> Chicago -> Cost = 689
Syracuse -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 920
Syracuse -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1077
Syracuse -> Buffalo -> Detroit -> Cleveland -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1252
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 908
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1039
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1146
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Chicago -> Cost = 844
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1001
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 951
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1199
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1330
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1437
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1713
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1135
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1292
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1242
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1518
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1405
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1562
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1512
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1410
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland ->

Chicago -> Cost = 1641
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1798
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1242
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1373
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1480
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1756
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1178
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1335
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1285
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1561
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1448
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1605
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1555
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1453
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1684
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1841
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1472
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1603
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1710
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1986
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1408
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1565
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1515
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1791
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1678
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland ->

Columbus -> Indianapolis -> Chicago -> Cost = 1835
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1785
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1683
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1914
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2071
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1515
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1646
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1753
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2029
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1451
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1608
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1558
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1834
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1721
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1878
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1828
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1726
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1957
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2114
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1488
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1619
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1726
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2002
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1424
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1581
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh ->

Cleveland -> Detroit -> Chicago -> Cost = 1531
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1807
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1694
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1851
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1801
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1699
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1930
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2087
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1531
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1662
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1769
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2045
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1467
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1624
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1574
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1850
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1737
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1894
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1844
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1742
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1973
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2130
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1101
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1232
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago

-> Cost = 1339
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1615
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1037
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1194
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1144
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1420
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1307
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1464
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1414
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1312
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1543
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1700
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1144
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1275
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1382
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1658
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1080
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1237
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1187
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1463
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1350
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1507
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1457
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1355
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1586
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1743

BFS paths:

Syracuse -> Buffalo -> Cleveland -> Chicago -> Cost = 684
Syracuse -> Buffalo -> Detroit -> Chicago -> Cost = 689
Syracuse -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 791
Syracuse -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 920
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Chicago -> Cost = 844
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1037
Syracuse -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 841
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 908
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1039
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 951
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1135
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1101
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1232
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1144
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1307
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1312
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1080
Syracuse -> Buffalo -> Cleveland -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1016
Syracuse -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1077
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1146
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1001
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1199
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1330
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1242
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1405
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1410
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1178

Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1408
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1339
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1194
Syracuse -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1420
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1414
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1543
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1144
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1275
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1187
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1350
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1355
Syracuse -> Buffalo -> Detroit -> Cleveland -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1252
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1437
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1292
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1518
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1512
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1641
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1242
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1373
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1285
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1448
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1453
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1472
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1603
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1515

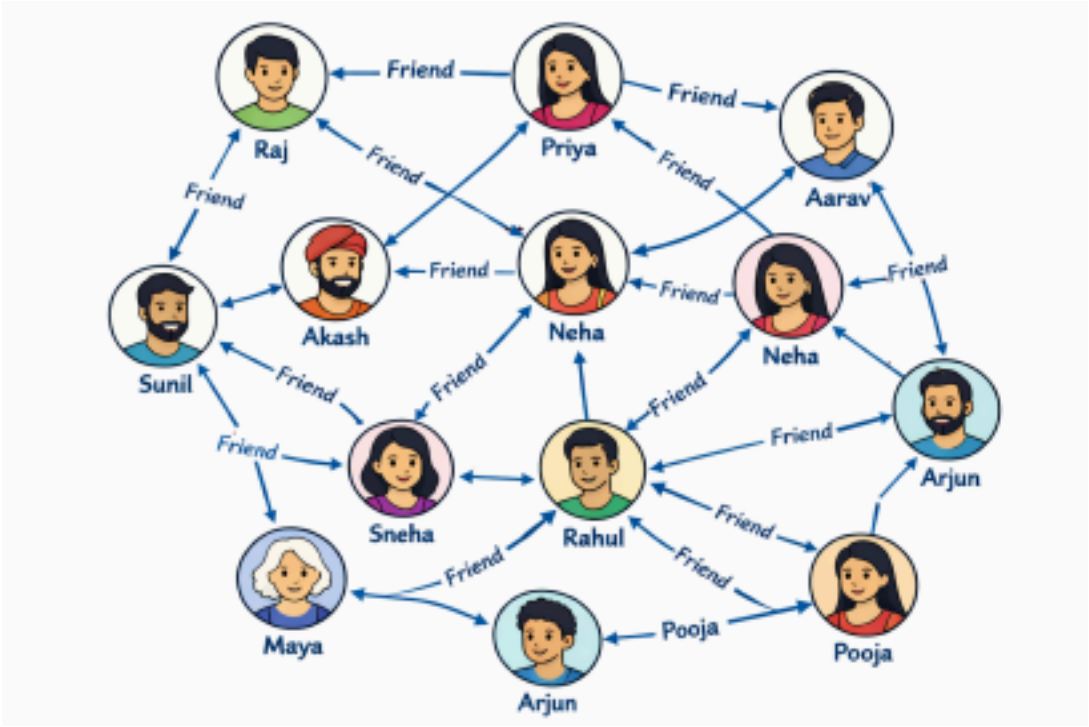
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1678
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1683
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1451
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1424
Syracuse -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1615
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1464
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1382
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1237
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1463
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1457
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1586
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1713
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1562
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1480
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1335
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1561
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1555
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1684
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1710
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1565
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1791
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1785
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1914
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1515
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1646

Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1558
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1721
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1726
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1488
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1619
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1531
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1694
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1699
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Chicago -> Cost = 1467
Syracuse -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1700
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1658
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1507
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1798
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1756
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1605
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1986
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1835
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1753
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1608
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1834
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1828
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1957
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1726
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1581
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1807

Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1801
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1930
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Indianapolis -> Chicago -> Cost = 1531
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Chicago -> Cost = 1662
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Detroit -> Chicago -> Cost = 1574
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Chicago -> Cost = 1737
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Chicago -> Cost = 1742
Syracuse -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1743
Syracuse -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1841
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2071
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2029
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1878
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2002
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1851
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago -> Cost = 1769
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1624
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 1850
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Detroit -> Chicago -> Cost = 1844
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago -> Cost = 1973
Syracuse -> Boston -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2114
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2087
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Columbus -> Cleveland -> Buffalo -> Detroit -> Chicago -> Cost = 2045
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 1894
Syracuse -> Boston -> Providence -> New York -> Philadelphia -> Baltimore -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago -> Cost = 2130

BFS Exploration Cost = 3342

Q2 Consider a social media network as given below. Find BFS and DFS tree.




```
#include <iostream>
using namespace std;




string name[12] = {
    "Raj", "Priya", "Aarav",
    "Sunil", "Akash", "Neha",
    "Sneha", "Rahul", "Maya",
    "Arjun", "Pooja", "Arjun2"
};

int graph[12][20];
int edgeCount[12];

void add_edge(int u, int v){
```

```
    graph[u][edgeCount[u]++] = v;
    graph[v][edgeCount[v]++] = u;
}

void make_graph(){
    for(int i = 0; i < 12; i++){
        edgeCount[i] = 0;
    }

    add_edge(0,1); // Raj - Priya
    add_edge(0,3); // Raj - Sunil
    add_edge(1,2); // Priya - Aarav
    add_edge(1,5); // Priya - Neha
    add_edge(2,5); // Aarav - Neha
    add_edge(2,9); // Aarav - Arjun
    add_edge(3,4); // Sunil - Akash
    add_edge(3,6); // Sunil - Sneha
    add_edge(4,5); // Akash - Neha
    add_edge(5,7); // Neha - Rahul
    add_edge(5,9); // Neha - Arjun
    add_edge(6,7); // Sneha - Rahul
    add_edge(6,8); // Sneha - Maya
    add_edge(7,9); // Rahul - Arjun
    add_edge(7,10); // Rahul - Pooja
    add_edge(8,11); // Maya - Arjun2
    add_edge(9,10); // Arjun - Pooja
    add_edge(10,11); // Pooja - Arjun2
}

void dfs(int u, int vis[]){
    vis[u] = 1;
    cout << name[u] << " ";

    for(int i = 0; i < edgeCount[u]; i++){
        int v = graph[u][i];
        if(!vis[v])
            dfs(v, vis);
    }
}

void bfs(int s){
```

```
    int queue[12];
    int front = 0, rear = 0;
    int vis[12] = {0};

    queue[rear++] = s;
    vis[s] = 1;

    while(front < rear){
        int u = queue[front++];
        cout << name[u] << " ";

        for(int i = 0; i < edgeCount[u]; i++){
            int v = graph[u][i];
            if(!vis[v]){
                vis[v] = 1;
                queue[rear++] = v;
            }
        }
    }
}

int main(){
    make_graph();

    int vis[12] = {0};

    cout << "DFS Tree:\n";
    dfs(0, vis);//Raj

    cout << "\n\nBFS Tree:\n";
    bfs(0);

    return 0;
}
```

DFS Tree:

Raj Priya Aarav Neha Akash Sunil Sneha Rahul Arjun Pooja Arjun2 Maya

BFS Tree:

Raj Priya Sunil Aarav Neha Akash Sneha Arjun Rahul Maya Pooja Arjun2